Hyperlocal Air Quality Mapping and AQI Prediction Around Educational Institutions Using Low-Cost Sensors and Machine Learning

SUBMITTED TO
BABU BANARASI DAS UNIVERSITY
LUCKNOW



IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF

MASTER OF COMPUTER APPLICATIONS

SUBMITTED BY
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(Session: 2026-27)

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1. INTRODUCTION

Air pollution is one of the main global health issues and leads to millions of deaths annually. Children and students are the most affected groups since their respiratory and brain systems are under development, their breathing rate is high, and they spend much time outside on the streets.

At present, air quality is measured in terms of reference stations; however, those stations are located quite far from each other (5-15 km). Thus, it means that measurements show only the general information about the pollution in the area but not any spikes in air pollution in specific zones, such as school drop off times, playground dust or construction.

To address this issue, Low-Cost Air Quality Sensors (LCAS), which use IoT, give an opportunity to measure air quality in real time and in street/campus zones. In combination with Machine Learning (ML), LCAS can correct the measurement and allow generating campus pollution maps and prediction of AQI.

2. MOTIVATION

Vulnerable Student Protection: Exposure to polluted air for a prolonged period results in asthma, decreased lung capacity, and poor academic outcomes in school students. There is a need for local data on campuses for protecting the well-being of students.

Addressing the Distance Gap Issue: The air quality index of the entire city does not indicate the air quality at school premises near a busy highway. Therefore, hyperlocal air quality data becomes necessary.

Cost-Effective Technology: While professional air quality monitoring stations require an investment of $50,000 or more, the IoT-based air quality sensing node requires less than $300 per unit.

Proactive Actions: If the school administration has knowledge about air quality index for the next 6-24 hours, they can take proactive actions by moving outdoor sports activities indoor, switching on air filters, or informing parents about asthmatic children.

3. BRIEF LITERATURE SURVEY

Recent findings (2024-2025) include advancements in three key areas:

ML for Sensor Calibration: Investigations confirm that data produced by low-cost particulate matter ($PM_{2.5}$) sensors may suffer from errors associated with excessive humidity and changes in temperature. Calibration of these devices using machine learning models such as Random Forest and Gradient Boosting is efficient in solving the issue (He et al., 2024; Kumar & Singh, 2024).

AQI Prediction Using Deep Learning Models: Current advanced deep learning techniques including Transformers and Spatio-Temporal GNN learn complicated patterns from historical pollution, weather and traffic data to predict air pollution trends in 12-24 hour timeframe (Wang et al., 2024; Chen et al., 2024).

Analysis of School-Level Air Pollution: New deployment of sensors in educational facilities confirmed that levels of air pollution at school gates in morning rush hour reach 3-5 times higher values compared to nearby neighbourhood that is not detected by standard urban monitoring systems (Patel et al., 2024).

4. GAPS IN THE STUDY

Not a Full Monitoring System for Campus: Almost all researches cover only the calibration part or city prediction but almost not a full system from the calibration to predictions made for educational campuses.

The Impact of Weather in the Tropical Climate: Calibration models were studied mostly in the mild climate and don't work well during the tropical weather conditions (such as high humidity and heat during the summer season).

Lack of Specific Campus Data for Prediction Models: Prediction models don't include any specific data regarding the educational campus such as the time of rush hours and playground schedule.

Lack of User-Friendly Tools: Research rarely translates complex ML predictions into simple, actionable dashboards that non-technical users (school principals, teachers, and parents) can easily understand.

5. PROBLEM FORMULATION

Given the limitations of existing air quality monitoring infrastructure in capturing hyperlocal variations around educational institutions, and the demonstrated vulnerability of student populations to pollutant exposure, this research addresses the following core problem:

How can a cost-effective, scalable network of low-cost air quality sensors, integrated with machine learning models, be designed and deployed to generate accurate, real-time hyperlocal AQI maps and reliable short-term (6–24 hour) AQI predictions for educational institution campuses and their immediate surroundings?

This problem decomposes into the following sub-problems:

- **P1:** How to select, calibrate, and validate LCAS nodes against reference-grade instruments under diverse environmental conditions to achieve acceptable accuracy ($R^2 > 0.85$, RMSE < 15 $\mu g/m^3$ for $PM_{2.5}$)?
- **P2:** How to optimally place sensors across a campus to maximize spatial coverage while minimizing hardware costs?
- **P3:** How to fuse multi-source data (sensor readings, meteorological variables, traffic data, satellite imagery) into a unified ML framework for spatiotemporal AQI prediction?
- **P4:** How to translate model outputs into an intuitive, real-time decision-support dashboard for non-technical end users (school administrators, parents, health officials)?

6. OBJECTIVES

Implement an IoT Sensor Network: Construct and deploy inexpensive sensor devices ($PM_{2.5}$, PM_{10}, NO_2, CO, temperature, humidity) in 3-5 educational establishments.

Train ML Algorithms for Calibration: Calibrate machine learning algorithms (Random Forest, XGBoost) on co-located data of government stations to provide highly accurate measurements ($R^2 > 0.85$).

Create Campus Heatmaps: Generate highly detailed heatmaps (50-100 m resolution) in real-time (updated once per 15 minutes) through spatial interpolation.

Design Decision-Support Platform: Create a user-friendly mobile/desktop application providing heatmaps, forecasts, and colour health warnings.

7. METHODOLOGY / PLANNING OF WORK

The proposed research will be carried out in several stages, starting with the collection of air-quality and environmental data from selected educational institutions. Low-cost IoT sensor nodes will be installed at suitable locations such as school gates, playgrounds, roadsides, and other areas where students are exposed to outdoor air pollution. The sensors will measure parameters such as $PM_{2.5}$, PM_{10}, temperature, and humidity at regular intervals. The collected data will be transmitted to a central server through Wi-Fi/Internet and stored in a database.

In the second stage, the collected sensor data will be cleaned and processed to remove missing, noisy, and abnormal observations. The low-cost sensor readings will then be calibrated using data from reliable reference monitoring stations. Machine learning algorithms such as Random Forest and XGBoost will be trained using pollutant and meteorological variables to improve the accuracy of the sensor measurements.

In the third stage, the calibrated measurements from different sensor locations will be used to generate hyperlocal air-quality maps. Spatial interpolation techniques will be applied to estimate pollution levels between sensor locations and identify pollution hotspots around educational campuses, particularly near traffic areas and school entrances.

In the fourth stage, historical air-quality, weather, and time-related data will be used to develop AQI forecasting models. Models such as XGBoost, LSTM, or CNN-LSTM will be evaluated for predicting AQI for 6, 12, and 24 hours ahead. The models will be compared using performance measures such as MAE, RMSE, MAPE, and R^2.

Finally, a user-friendly web-based dashboard will be developed to display real-time AQI, pollutant levels, campus heatmaps, historical trends, and future AQI predictions. The system will also provide simple health warnings and recommendations for school administrators. The overall performance, accuracy, cost, and scalability of the proposed framework will be evaluated using the collected field data.

8. EXPECTED OUTCOMES

Accurate, Low-Cost Monitoring: A field-tested sensor node design costing under $300 that matches professional station accuracy within ±15%.

High-Resolution Campus Heatmaps: Visual 50–100 meter maps showing exact pollution hotspots across school grounds updated every 15 minutes.

Reliable AQI Predictions: Accurate 6- to 24-hour advance warnings for air quality spikes (MAPE < 15%).

Actionable School Dashboard: A simple digital tool that gives teachers and administrators clear guidance on when to adjust outdoor activities.

Practical Policy Evidence: Concrete data identifying high-risk zones (e.g., school drop-off gates), supporting policies like school-zone traffic restrictions and green buffer zones.

Academic Publications: 2 to 3 peer-reviewed research papers documenting the calibration, mapping, and forecasting methodologies.

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